



Postdoctoral Researcher Position in Agentic AI for Hardware Design



A Joint University of Minnesota Twin Cities × Georgia Tech Postdoctoral Opportunity

Prof. Caiwen Ding

University of Minnesota Twin Cities

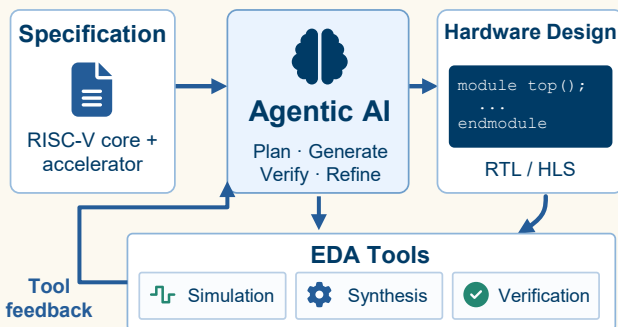
Prof. Yingyan (Celine) Lin

Georgia Tech

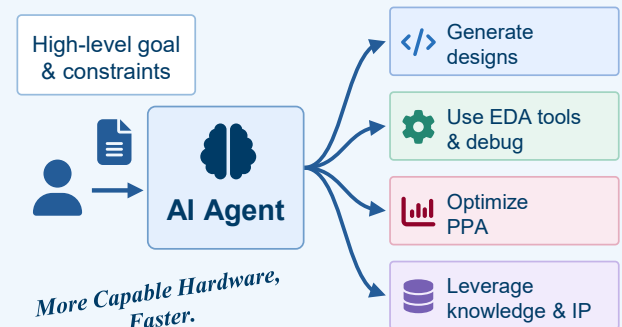
Hardware design remains expertise-intensive and iteration-heavy. This joint postdoctoral position will focus on developing foundation-model-powered AI agents that move beyond isolated code generation toward **autonomous, tool-integrated hardware-design workflows**. These agents can interpret specifications, reason about design choices, generate hardware implementations, and use tool feedback to diagnose and repair design errors.

The successful candidate will be co-mentored by both PIs, with Prof. Lin as the primary mentor, and will work closely with researchers in both groups at the intersection of **agentic AI, electronic design automation (EDA), computer architecture, and efficient computing systems**.

From Specification to Working Hardware



Toward Autonomous Hardware Designers



Research Focus

- Agentic hardware design from specifications.** LLM-based and multi-agent systems for specification understanding, requirement decomposition, planning, and scalable RTL/HLS generation.
- Tool-integrated verification and debugging.** Use simulation, synthesis, and verification feedback to identify design errors, reason about their causes, and iteratively repair and refine designs.
- Hardware-aware optimization and design-space exploration.** Explore and optimize architectures and implementations using correctness, PPA (power, performance, area), and timing feedback.
- Knowledge- and repository-augmented hardware agents.** Leverage HDL repositories, design documentation, reusable components, and domain knowledge while coordinating specialized agents on complex hardware-design tasks.

What You'll Do

- Build agent frameworks, benchmarks, and evaluation infrastructure for realistic hardware-design tasks and toolchains.
- Publish research, collaborate across both groups, and mentor students.

Who You Are

- Ph.D. in CS, CE, EE, or a related field, completed by the start date.
- Strong research record, demonstrated ability to conduct independent research, and strong programming/system-building skills.
- Expertise in either LLMs/AI agents/ML systems or hardware design, EDA, computer architecture, or hardware/software co-design.
- Interest in applying AI to hardware design, with strong communication and collaboration skills.

Apply

Please email these materials to both PIs:

- Cover letter
- CV, including a publication list
- Brief research statement: prior work and future interests
- 2–3 representative publications (PDFs or links)
- Names and contact details of three references; letters may be requested later

✉ **Prof. Caiwen Ding**
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✉ **Prof. Yingyan (Celine) Lin**
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Rolling review until filled. Candidates from AI/LLM-agent or hardware/EDA backgrounds are encouraged to apply.